

Statistical Tests

significant isn't the same as important

Why run a test at all?

- You already eyeball your data for patterns. Tests just put a number on what you suspect.
- A test doesn't replace your judgment. It tells you whether a pattern is probably real, not whether it matters.
- “Statistically significant” is a math term. It is not a synonym for “important,” “large,” or “newsworthy.”
 - With enough rows, even tiny, meaningless differences can turn up “significant.”

Meet the dataset

- 118 rows, one per school: district, school, number_of_students, pcttest, pctpoor, avg_score, ZIP
- pcttest = share of students tested · pctpoor = share qualifying as low-income · avg_score = average test score
- Before we test anything: sort, filter and skim it the way you would any new dataset.
- Make a list of the problems

```
district, school, number_of_students, pcttest, pctpoor, avg_score, ZIP
```

Before you test: validate the ranges

- pcttest and pctpoor are percentages: where should every value fall?
- This file has pctpoor values of 120 and 300, a pcttest value of 200, and one row where pcttest reads literally “yes.”
- Run a range-check column before anything else:

```
=OR(D2<0, D2>100, E2<0, E2>100)
```

returns TRUE for every value a test should not see

4 of 118 rows should get dropped before the analysis – go ahead and delete them

Before you test: check the frame itself

- Zip code range?
- Where is “WENTZVILLE MIDDLE”, exactly?
- Neither of these breaks a formula. Both should stop you before you trust what a district-level test is actually measuring.

```
=COUNTIF(B:B, B2)>1 flags school names that appear more than once
```

Start with description, not a test

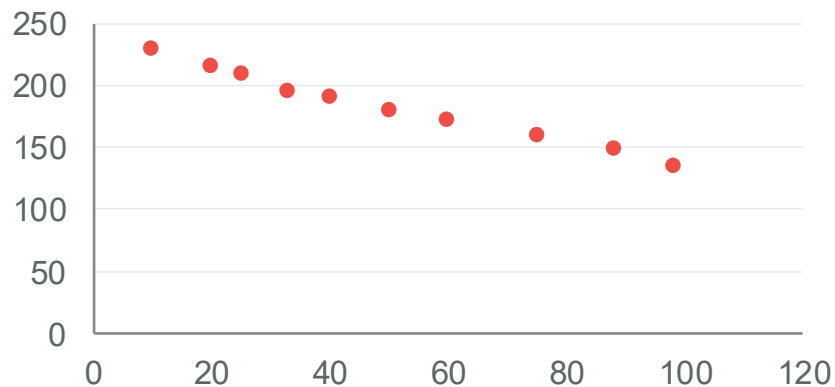
- Once the 4 bad rows are excluded, 114 rows remain — 112 of them have scores.
- AVERAGE and MEDIAN together tell you whether a distribution is skewed before you do anything fancier.

	mean	median
avg_score	190.3	194.8
pctpoor	44.5	33.8

pctpoor's mean (44.5) sitting well above its median (33.8) already signals a right-skewed distribution: a few high-poverty schools are pulling the average up.

Correlation: CORREL()

- Question: as a school's poverty rate rises, does its average score move with it?
- `=CORREL(E2:E115, F2:F115)` returns $r = -0.62$. CORREL skips the two blank-score rows automatically.
- r ranges from -1 to 1 . -0.62 is a fairly strong negative relationship, that poorer schools tend to score lower.



illustrative shape, not the real plot

r tells you two variables move together. It never tells you poverty causes lower scores. Funding, staffing and test-prep resources all move with poverty too.

Setup: delete the 4 bad rows as whole rows first. If you sort, sort entire rows, never one column – CORREL doesn't care about order, but each pctpoor must stay on the same row as its school's score.

What a test does — and doesn't — tell you

- A test confirms a pattern is probably real. It never confirms the pattern is important, fair, or fully explained.
- Maybe the most useful finding on this whole file (one shared ZIP, duplicated school names) wasn't a test at all. Validate before you calculate.
- Low test-participation rates concentrated in high-poverty schools are their own story — absence of data is data.